

```
export/zfs/bonwick
```

Note that the user **bonwick** only has 10 GB of space available, while the user **billm** can use the full pool (67 GB).

Now you have learned some basic commands to create and manage a ZFS **zpool** and file system. But it will be interesting to know what are the difference between the traditional file systems and ZFS file systems. In the following section, we would bring out some differences which ZFS has with other file systems.

4.8.4 ZFS file system granularity

Historically, file systems have been constrained to one device so that the file systems themselves have been constrained to the size of the device. Creating and re-creating traditional file systems because of size constraints are time-consuming and sometimes difficult. Traditional volume management products helped manage this process. Because ZFS file systems are not constrained to specific devices, they can be created easily and quickly, similar to the way directories are created. ZFS file systems grow automatically within the space allocated to the storage pool. Instead of creating one file system, such as **/export/home**, to manage many user subdirectories, you can create one file system per user. In addition, ZFS provides a file system hierarchy so that you can easily set up and manage many file systems by applying properties that can be inherited by file systems contained within the hierarchy.

4.8.5 ZFS space accounting

ZFS is based on a concept of pooled storage. Unlike typical file systems, which are mapped to physical storage, all ZFS file systems in a pool share the available storage in the pool. So, the available space reported by utilities such as **df** might change even when the file system is inactive, as other file systems in the pool consume or release space. Note that the maximum file system size can be limited by using quotas. Space can be guaranteed to a file system by using reservations. This model is very similar to the NFS model, where multiple directories are mounted from the same file system (consider **/home**).

All metadata in ZFS is allocated dynamically. Most other file systems pre-allocate much of their metadata. As a result, an immediate space cost at file system creation for this metadata is required. This behavior also means that the total number of files supported by the file systems is predetermined.